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General Training for Programming Competitions & Coding Interviews

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Linked Lists, Stacks, Queues, Binary Trees and Tries

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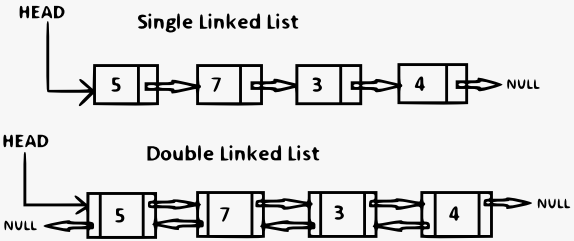
Linked Lists

Linked Lists Basics

- A **(singly) linked list** is a data structure that contains a sequence of nodes.
- Each node includes an object and a reference to the next node (if any).
- The first node is referred to as the **head** and the last node is referred to as the **tail**.
- In a **doubly linked list**, each node also has a link to its predecessor.



Operations on linked lists

- **Inserting** or **deleting** nodes locally has $\mathcal{O}(1)$ time complexity.
- **Searching** for an element takes $\mathcal{O}(n)$ time.
- Unlike an array, a linked list does not provide **constant time access** to a particular "index" within the list.
- You should have "good reasons" to use a linked list instead of an array in your code.

Delete a Given Node From a Singly Linked List

Given a node in a singly linked list, **deleting it** in $\mathcal{O}(1)$ time appears impossible because its predecessor's next field has to be updated. However, there is a trick that can achieve this if the input node is guaranteed not to be the tail node.

Delete a Given Node From a Singly Linked List

Given a node in a singly linked list, **deleting it** in $\mathcal{O}(1)$ time appears impossible because its predecessor's next field has to be updated. However, there is a trick that can achieve this if the input node is guaranteed not to be the tail node.

delete a node without knowing its predecessor

```
1 // assume node_to_delete is not tail
2 auto successor = node_to_delete->next;
3 // copy (or move) data from successor
4 node_to_delete->data = successor->data;
5 // copy next node from successor
6 node_to_delete->next = successor->next;
7 // delete successor completely from memory
8 delete successor;
```

If we copy the **value part of the next node** to the current node, and then **delete the next node**, we have effectively deleted the current node. The time complexity is $\mathcal{O}(1)$.

The Runner Technique

The **runner** (or second pointer) **technique** is used in many linked list problems.

- Idea: Use two pointers that traverse the list simultaneously.
- The fast pointer could be ahead with a fixed offset, or could use multiple steps each time the slow pointer moves one.
- (Compare this to the techniques we used to find & delete elements from a string/array.)

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Task: Find the midpoint of a linked list in a single iteration.

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Task: Find the midpoint of a linked list in a single iteration.

Solution:

- One pointer moves twice as quickly as the other.
- When the fast pointer reaches the end, the slow pointer is in the middle.
- In practice: Handle lists of even length safely.

Test for Cyclicity

Write a program that takes the head of a singly linked list and finds out whether there is a **cycle** in the list.
(Requirements: $\mathcal{O}(1)$ space, $\mathcal{O}(n)$ time)

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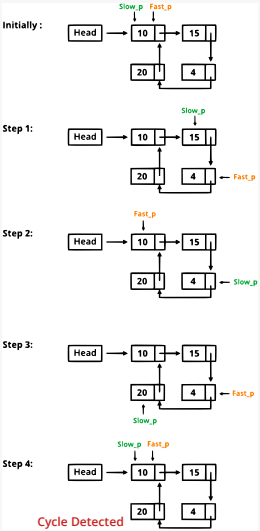
has_cycle.cpp

```cpp
bool has_cycle(Node *node) {
  Node *slow = node;
  Node *fast = node;
  while (fast && fast->next) {
    // advance slow by one
    slow = slow->next;
    // advance fast by two
    fast = fast->next->next;
    if (slow == fast) {
      return true;
    }
  }
  return false;
}
```



Test for Cyclicity

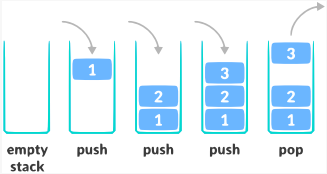
This works, because when the the slow pointer enters the cycle it is in **front of the fast pointer**. Since the fast pointer moves twice as fast, the distance between them is reduced by 1 in each step, until they meet.
Therefore they meet while the slow pointer traverses the cycle for the first time.

If the fast pointer reaches `null`, there is no cycle!

Stacks & Queues

Stacks Basics

- A stack uses **LIFO** (last-in, first-out) ordering. That is, as in a stack of dinner plates, the item that was **added last** is **removed first**.



- A **stack can be implemented** with a **linked list** or a dynamically **resizable array**.
- Simple **parsing problems** can often be implemented with a stack.
  (Learn to recognize problems when the stack LIFO property is applicable.)
- A stack can also be used to **implement a recursive algorithm iteratively**.
- **Consider using the C++ stack class** instead of implementing your own stack class.
  https://www.geeksforgeeks.org/stack-in-cpp-stl/

A `std::vector` Can Be Also Used as a Stack

vector_stack.cpp

```cpp
// A vector can be used as a stack.
// This (unnecessary) wrapper demonstrates how.
template <typename T> class Stack {
private:
  std::vector<T> data;

public:
  void push(T item) { this->data.push_back(item); }
  bool empty() { return this->data.empty(); }
  T top() { return this->data.back(); }
  void pop() { this->data.pop_back(); }
};
```

A `std::vector` semantically **offers stack functionality**. Using a vector directly as a stack **can speed up code**.

Stack in Python

In python you can also use a **list** as an easy stack implementation.

```
# initialize stack as a list:
stack = []
# push an element to the end of the stack:
stack.append(5)
# check if stack is empty:
len(stack) == 0
# get topmost (i.e. last) element:
stack.pop(-1)
```

However, this is **not the fastest way in python** if the stack becomes large. In the case of large stacks, better use **Collections.deque** or **queue.LifoQueue** (see https://www.geeksforgeeks.org/stack-in-python/).

Test a String Over {}()[] for Well-Formedness

A string over the characters {}()[] is said to be **well-formed if** the **different types of brackets match in the correct order**. For example, ([]){()} is well-formed. However, {) and [()[]{()() are not well-formed. Write a program that tests if a string made up of the characters {}()[] is well-formed.

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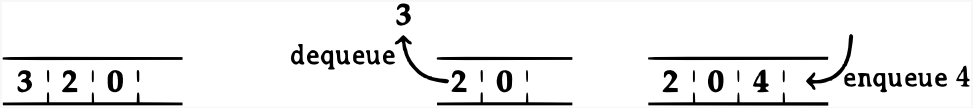
is_well_formed.cpp

```
1
2  bool is_well_formed(const string &s) {
3    stack<char> chars;
4    for (const char c : s) { // iterate over all characters in s
5      if (c == '(' || c == '{' || c == '[') {
6        chars.push(c);
7      } else {
8        // missing corresponding left bracket
9        if (chars.empty()) { return false; }
10       if ((c == ')' && chars.top() != '(') ||
11           (c == '}' && chars.top() != '{') ||
12           (c == ']' && chars.top() != '[')) {
13         return false;
14       }
15       chars.pop();
16     }
17   }
18   // if stack not empty we have unmatched left brackets
19   return chars.empty();
```

Queues Basics

- Add (**enqueue**) and remove (**dequeue**) elements in **FIFO** (first-in, first-out) order.

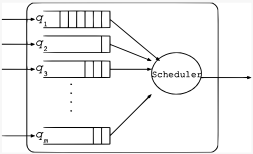


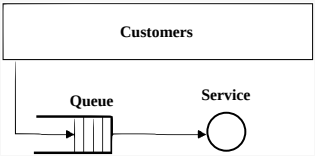
- A **queue can be implemented efficiently** using a **linked list** or a (dynamically resizable circular) **array**.

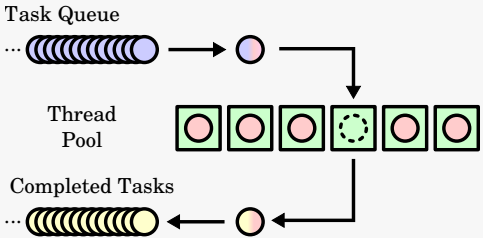
Queues in practice

queue_with_list.cpp

```cpp
// simple queue implementation with a list
template <typename T> class Queue {
private:
  list<T> data;

public: // a std::list can be used as a queue
  void enqueue(T x) { data.emplace_back(x); }

  T dequeue() {
    const T val = data.front();
    data.pop_front();
    return val;
  }

  // convenience functions
  size_t size() { return data.size(); }
  bool empty() { return data.empty(); }
```







Binary Trees & Recursion

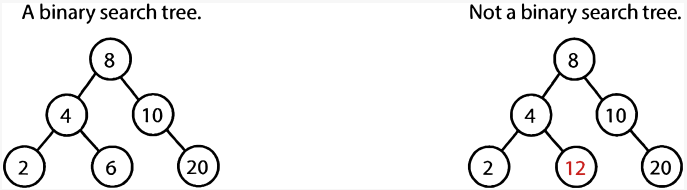
Binary Trees Basics

A tree is a **type of graph**. (We will cover general graphs later.)

- Each tree has a **root node**.
- The root and each other node has zero or more **child nodes**. Nodes without children are also called **leaf nodes**.
- The tree **must not contain cycles**. The nodes may or may not be in a particular order, they could have any data type as values, and they may or may not **have links back to their parent nodes**.
- The **height** of a tree is the length of the longest path from the root to a leaf.

Binary Trees Basics

A binary tree is a tree in which each node **has up to two children**. A binary search tree imposes the condition that, for each node, its **left descendents are less than or equal to** the current node, **which is less than the right descendents**.

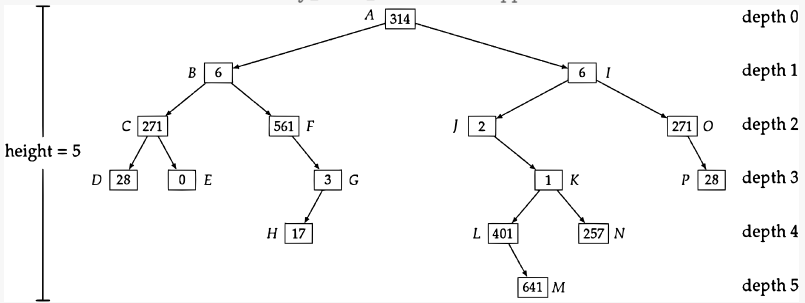


Special kinds of balanced binary search trees are the **Red-Black tree**, and the **AVL tree**.

Binary Tree Traversal

You should be comfortable implementing **in-order**, **post-order**, and **pre-order** traversal.

binary_tree_traversal.cpp



pre-order

```
// A B C D E F G H I J K L M N O P
void pre_order(Binary_Tree_Node *node) {
  if (node) {
    print(node); // process node here
    pre_order(node->left);
    pre_order(node->right);
  }
}
```

in-order

```
// D C E B F H G A J L M K N I O P
void in_order(Binary_Tree_Node *node) {
  if (node) {
    in_order(node->left);
    print(node); // process node here
    in_order(node->right);
  }
}
```

post-order

```
// D E C H G F B M L N K J P O I A
void post_order(Binary_Tree_Node *node) {
  if (node) {
    post_order(node->left);
    post_order(node->right);
    print(node); // process node here
  }
}
```

Binary Tree Traversal Without Recursion

Often it is more convenient to solve **binary tree problems** by traversing the tree **with recursion**. But using a stack, we can implement tree traversals without recursion.

binary_tree_traversal_no_recursion.cpp

pre-order without recursion

```
void pre_order_no_recursion(Binary_Tree_Node *node) {
  stack<Binary_Tree_Node *> s;
  s.push(node);
  while (!s.empty()) {
    node = s.top();
    s.pop();
    if (node != nullptr) {
      print(node); // process node here
      // add left child after right child, since we
      // want to continue with the left child
      s.push(node->right);
      s.push(node->left);
    }
  }
}
```

in-order without recursion

```
void in_order_no_recursion(Binary_Tree_Node *node) {
  stack<Binary_Tree_Node *> s;
  while (!s.empty() || node) {
    if (node) {
      s.push(node);
      node = node->left; // going left
    } else {
      node = s.top(); // going up
      s.pop();
      print(node);        // process node here
      node = node->right; // going right
    }
  }
}
```

Recursion Elimination

Solving problems by recursion is often the intuitive way. However, recursive functions can be bad for the performance of your code:

- All recursive calls require memory.
- Recursive calls are managed by a stack, this can overflow if the recursion is too deep.
- For example, python has a *RecursionError: maximum recursion depth exceeded* error to prevent the stack overflow.

useless_recursion.py

```
def useless_recursion(n):
   if n == 0:
      print ("done.")
      return
   else:
      return useless_recursion(n-1)

if __name__ == "__main__":
   print ("case 990:")
   useless_recursion(990)
   print ("case 1000:")
   useless_recursion(1000)
```

Recursion Elimination

Try to eliminate your recursion, or transform it into a tail-recursion!

Bad recursion:

recursion.cpp

```cpp
int factorial_bad(unsigned int a){
   // compute the factorial a! using a recursive function
   if (a == 0){
      return 1;
   }
   return a * factorial_bad(a-1);
}
```

- There is another operation happening *after* the recursive call!

Good (tail-) recursion:

recursion.cpp

```cpp
int factorial_tail(unsigned int a, unsigned long int ret=1){
   // compute the factorial a! using a tail-recursive function
   if (a < 2){
      return ret;
   }
   return factorial_tail(a-1, ret*a);
}
```

- Recursive call is the *last* operation.
- Can be automatically transformed into while-loop by compiler!

Recursion Elimination

Computing the n-th **Fibbonacci number** is the prime example of recursive algorithms:

Fibbonacci recursive

```
1 unsigned long int fib_rec(unsigned int n){
2    if (n<=2){
3       return 1;
4    }
5    return fib_rec(n-1)+fib_rec(n-2);
6 }
```

There is an obvious non-recursive way to compute the Fibbonacci number by a for-loop. But for the sake of this exercise, let us try to write this function as a tail-recursion.

Recursion Elimination

Computing the *n*-th **Fibbonacci number** as a tail recursion:

Fibbonacci tail-recursive

```
1  unsigned long int fib_tailrec(unsigned int n, unsigned int k=1, unsigned int last=1, unsigned int seclast=1){
2     // n : target fibonacci number
3     // k : current fibonacci number
4     // last, seclast: last and second-to-last numbers
5
6     if (n <= 2){
7        return 1;
8     }
9     if (k+1 == n){
10       return last;
11    }
12    return fib_tailrec(n, k+1, last+seclast, last);
13 }
```

Recursion Elimination

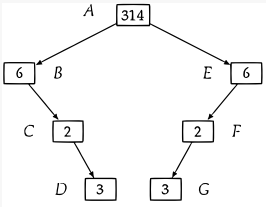
Depending on the optimization flag `-O`, `g++` can detect and eliminate some recursions automatically.

Experiments: compile and run `fibbo.cpp` and `recursion_squares.cpp` with different optimization flags (`-O0`, `-O1`, `-O2`).

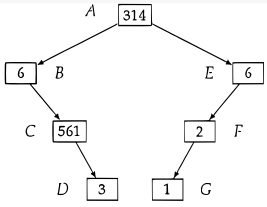
But *better save than sorry*: try to write your code directly without recursion, don't rely on the compiler!

Test if a Binary Tree Is Symmetric

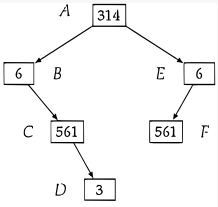
A binary tree is **symmetric** if you can draw a vertical line through the root and then the left sub-tree is the mirror image of the right sub-tree.



A symmetric binary tree.



An asymmetric binary tree.



An asymmetric binary tree.

Test if a Binary Tree Is Symmetric

Write a **program that checks** whether a **binary tree is symmetric**

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Write a **program that checks** whether a **binary tree is symmetric**

is_symmetric.cpp

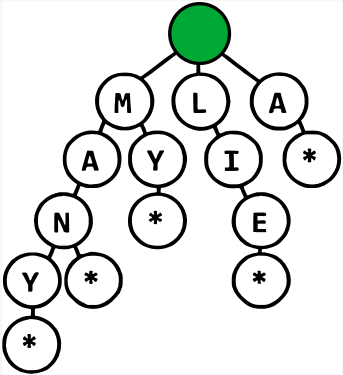
```cpp
bool check_symmetric(Binary_Tree_Node *subtree_left, Binary_Tree_Node *subtree_right) {
  if (subtree_left == nullptr && subtree_right == nullptr) {
    return true;
  } else if (subtree_left != nullptr && subtree_right != nullptr) {
    return subtree_left->data == subtree_right->data &&
           check_symmetric(subtree_left->left, subtree_right->right) &&
           check_symmetric(subtree_left->right, subtree_right->left);
  }
  return false; // one subtree is empty, and the other is not
}

bool is_symmetric(Binary_Tree_Node *node) {
  return node == nullptr || check_symmetric(node->left, node->right);
}
```

Prefix Trees ("Tries")

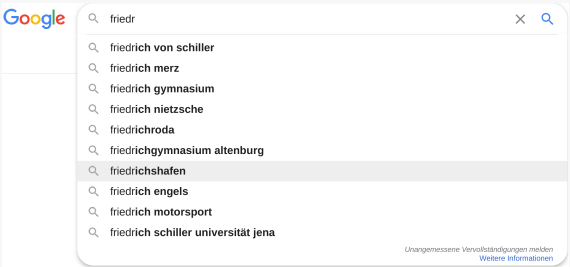
Prefix Trees ("Tries")

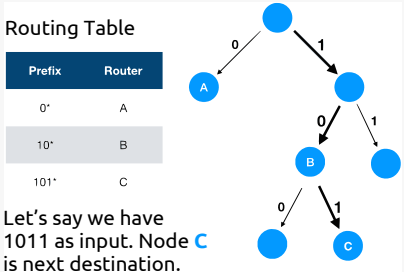
- A trie comes up a lot in interview questions, but algorithm textbooks don't spend much time on this data structure.
- The word **trie** is an infix of the word "re**trie**val".
  - A trie is a variant of an n-ary tree in which **characters are stored at each node**. **Each path** down the tree **may represent a word**.
  - The *** nodes** (sometimes called "null nodes") are often used to **indicate complete words**.

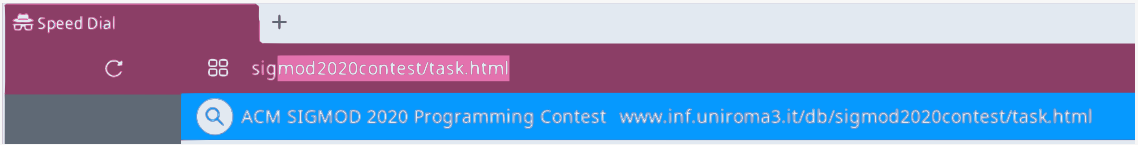


Where Are Tries Used?

Very commonly, tries are used to store entire dictionaries for quick **prefix lookups**. While a hash table can quickly look up whether a string is a valid word, it cannot tell us if a string is a prefix of any valid words. A trie can do this in $\mathcal{O}(K)$ time, where K is the length of the string.

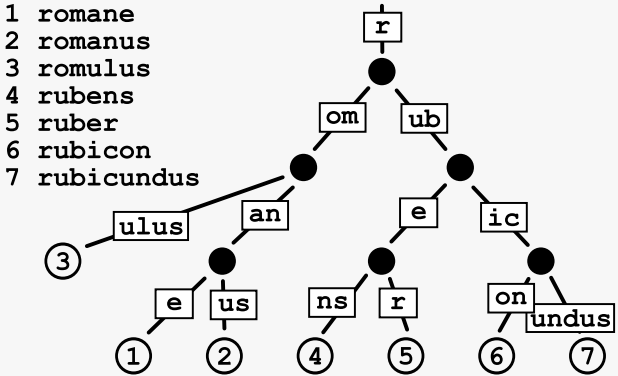






Radix Trees

A **space-optimized** (compressed) trie in which each node that is the only child is merged with its parent is called a **radix tree**.
(Read about **radix trees in databases**: https://db.in.tum.de/~leis/papers/ART.pdf)



Exercises (Upload Solutions to your Repository)

1. **Check if** a **linked list** is a **palindrome**.

   `is_palindrom.cpp` 

2. Write a program to **evaluate arithmetic expressions** in **Reverse Polish Notation**, for example a term like:

   `5 4 +`.

   `eval_rpn.cpp` 

3. **Implement** a **queue** by using **two stacks**.

   `queue_with_stacks.cpp`

4. **Implement** a function to **check if** a **binary tree is** a **binary search tree**.

   `is_binary_search_tree.cpp` 

5. **Find** the **shortest prefix** of a **string** which is **not a prefix of any string in the trie**.

   `shortest_prefix.cpp` 

Exam Questions

- **How** to **delete** a (non-tail) **node** from a **singly linked list without knowing its predecessor** in $\mathcal{O}(1)$ time?
- Explain the **runner technique** using a linked list **example**.
- If you were to **tune** the **code for speed**, **how would you implement a stack**?
- **How** can we **use a** `std::list` in C++ **as a queue**?
- **How** does a **binary search tree** differ from a **binary tree**?
- **Describe** the **three types** of **binary tree traversals**.
- **How** does a **radix tree** differ from a **trie**?